# Jay Doshi

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## Education

**B.Sc. in Computer Science & Engineering** **GPA: 3.7/4.0**
University of Toledo, Toledo, OH Expected Graduation: December 2027

**Relevant Coursework:** Statistics, Kalman Filtering, Database Systems, Operating Systems, Software Engineering

## Experience

**Network Engineer Intern** | Cenovus Energy, Oregon, OH **September 2026 – Present**

- Collaborate with cross-functional teams to validate system integrations, debug connectivity defects, and support deployments.
- Analyze SolarWinds NPM metrics for root-cause analysis of performance degradation, resolving issues before production outages.
- Integrate server, VMware, and backup systems into stable hosting environments, ensuring high availability for plant applications.

**Process Controls Intern** | Cleveland-Cliffs, Coatesville, PA **May 2026 – August 2026**

- Automated ETL pipelines moving live sensor data from legacy OpenVMS into SQL Server, cutting manual extraction by 40%.
- Engineered threshold-based anomaly detection with real-time email alerts to operators, reducing defective plate escapes by 30%.
- Shipped VB.NET journaling microservices to a clustered QMS app via GitHub, restoring PLC-to-SQL TCP/IP data flow in seconds.

**IT Support Technician Level 2** | University of Toledo, Toledo, OH **September 2025 – May 2026**

- Resolved 40+ monthly IT tickets by troubleshooting Windows, hardware, software, and network issues, minimizing downtime.
- Imaged and deployed 150+ Windows devices using CLI tools for BIOS updates, data backups, and software installation.

**Undergraduate Research Assistant** | University of Toledo, Toledo, OH **May 2025 – December 2025**

- Improved control accuracy 60% with Python scripts analyzing flight-test data from 15+ hardware trials to surface failure patterns.
- Reduced target navigation error by 40% by building and tuning a custom PID flight controller tested in a ROS2-Gazebo digital twin.
- Engineered a ROS2-Gazebo simulation pipeline to validate PX4 control logic on a digital twin in a physics virtual environment.
- Prototyped PID control designs in MATLAB, trialing gain settings for the PX4 drone flight controller.

## Professional Projects

**PolicyRAG Document Intelligence** | Python, GPT-4o, RAG, FAISS, FastAPI, Streamlit **December 2025 – January 2026**

- Built a RAG pipeline over 5 insurance policy documents (500+ pages), extracting text and tables from PDFs using pdfplumber.
- Indexed chunks in a FAISS vector store using OpenAI embeddings, reranking by cosine similarity to feed GPT-4o the top 5 clauses.
- Deployed a FastAPI endpoint with token auth answering batched queries in 40–50s; benchmarked retrieval latency and accuracy.

**DeepTruth** | Python, React, AWS, Gemini API, PyTorch, DistilBERT, MongoDB, GCP **January 2025 – March 2025**

- Won RocketHacks 2025 for developing scalable AI fact-checking product used by 1,000+ users over a 6-month period.
- Fine-tuned DistilBERT on 9.5K+ labeled records, enabling dual-model scoring with 70% Gemini and 30% BERT weighting.
- Designed AWS Lambda and CloudWatch cron workflows for scheduled Transformer retraining and ML lifecycle management.
- Automated CI/CD deployment via GitHub Actions and GCP VMs, reducing manual release steps and improving consistency.
- Built Python ETL pipeline with automated validation checks, improving data ingestion accuracy across 9,500+ records.

**SkillBridge Learning Platform** | Django, PostgreSQL, Gemini API, NLP, TF-IDF, AWS **August 2024 – December 2024**

- Developed Gemini API-powered AI agent for auto-generating video transcripts and MCQs, reducing manual content creation effort.
- Designed PostgreSQL schemas with dimensional modeling, improving query performance and reporting speed by 80%.
- Implemented JWT and Google OAuth authentication system, securing REST API endpoints and improving integration reliability.
- Reduced response time to under 15 seconds by designing microservices architecture integrating varied technologies across features.

## Technical Skills

**Languages & Frameworks**: Python, JavaScript, SQL, React, Django, FastAPI, Java, C++, C

**AI & Machine Learning**: LLM, RAG, Embeddings, Fine-Tuning, Transformer Models, NLP, LangChain, MCP, Agentic-AI Workflow, Hugging Face

**Databases & Cloud**: PostgreSQL, MongoDB, MySQL, Firebase, AWS (Lambda, S3, EC2, CloudWatch), Google Cloud Platform

**Software Development**: REST APIs, System Design, Data Structures & Algorithms, CI/CD, Unit Testing, Agile Methodologies, SCRUM

**Developer Tools**: Git, GitHub Actions, Docker, Linux Command Line, VS Code, Jupyter Notebook, Claude Code